

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: power.kicad_sch

3. MCU

STM32L431RC, SWD, status LED, fault pull-ups

File: mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 4-wire (ADS1220)

File: measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: rs485.kicad_sch

DOTECH HMT500 – Oil Moisture Transmitter – Schematic v0.2

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-30 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

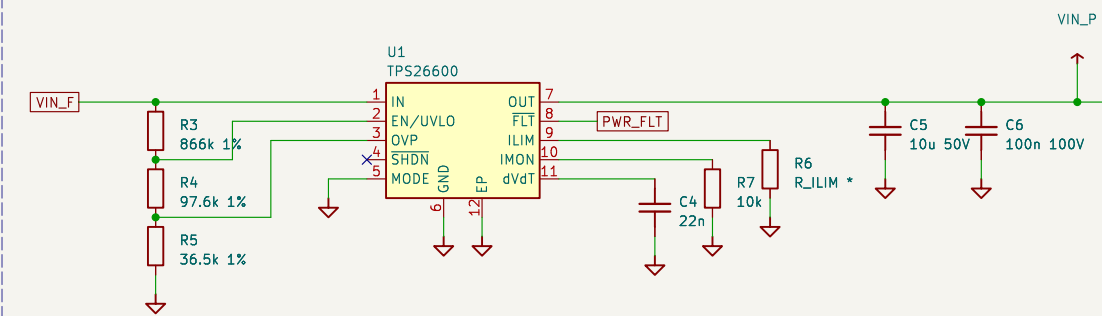
Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), VDDA, CHASSIS. Inter-sheet signals: global labels.

VERIFY: TPS2660, LMR36006, PCAP04, DAC8760, TPS26611 pin numbers are placeholders (datasheets not accessible when drawn).

TBD footprints: M Connect 8P, CMC, GDT, feedthrough, buck inductor, the five VERIFY ICs.

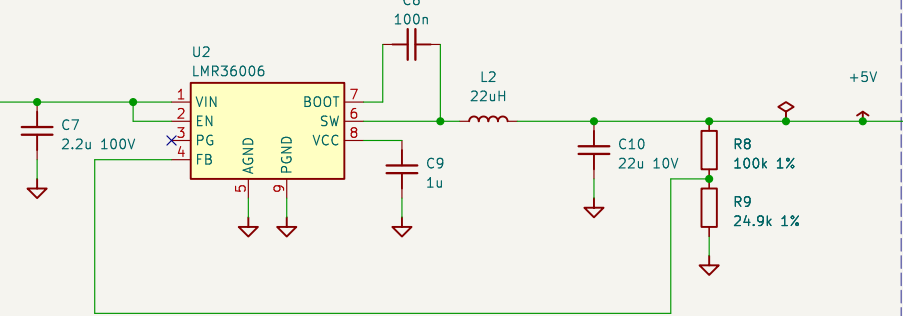
VERIFY=YES parts: placeholder pin numbers (TPS2660, LMR36006, PCAP04, DAC8760, TPS26611) Design notes: docs/hw/circuit-design.md Oil moisture transmitter HMT500 – RS-485 + 2x V/I analog output DOTECH Co., Ltd.		
Sheet: / File: HMT500.kicad_sch		
Title: HMT500 Top		
Size: A3	Date: 2026-09-26	Rev: 0.2
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/7

eFuse TPS2660 reverse -60 V / OVP 33 V / UVLO 9 V



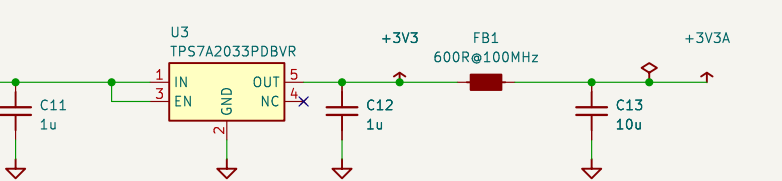
$UVLO = 1.2\text{ V} \times (R3+R4+R5)/(R4+R5) = 8.9\text{ V}$
 $OVP = 1.2\text{ V} \times (R3+R4+R5)/R5 = 32.9\text{ V}$ (Vref 1.2 V: VERIFY)
* R6 sets current limit 150 mA (formula in TPS2660 datasheet)

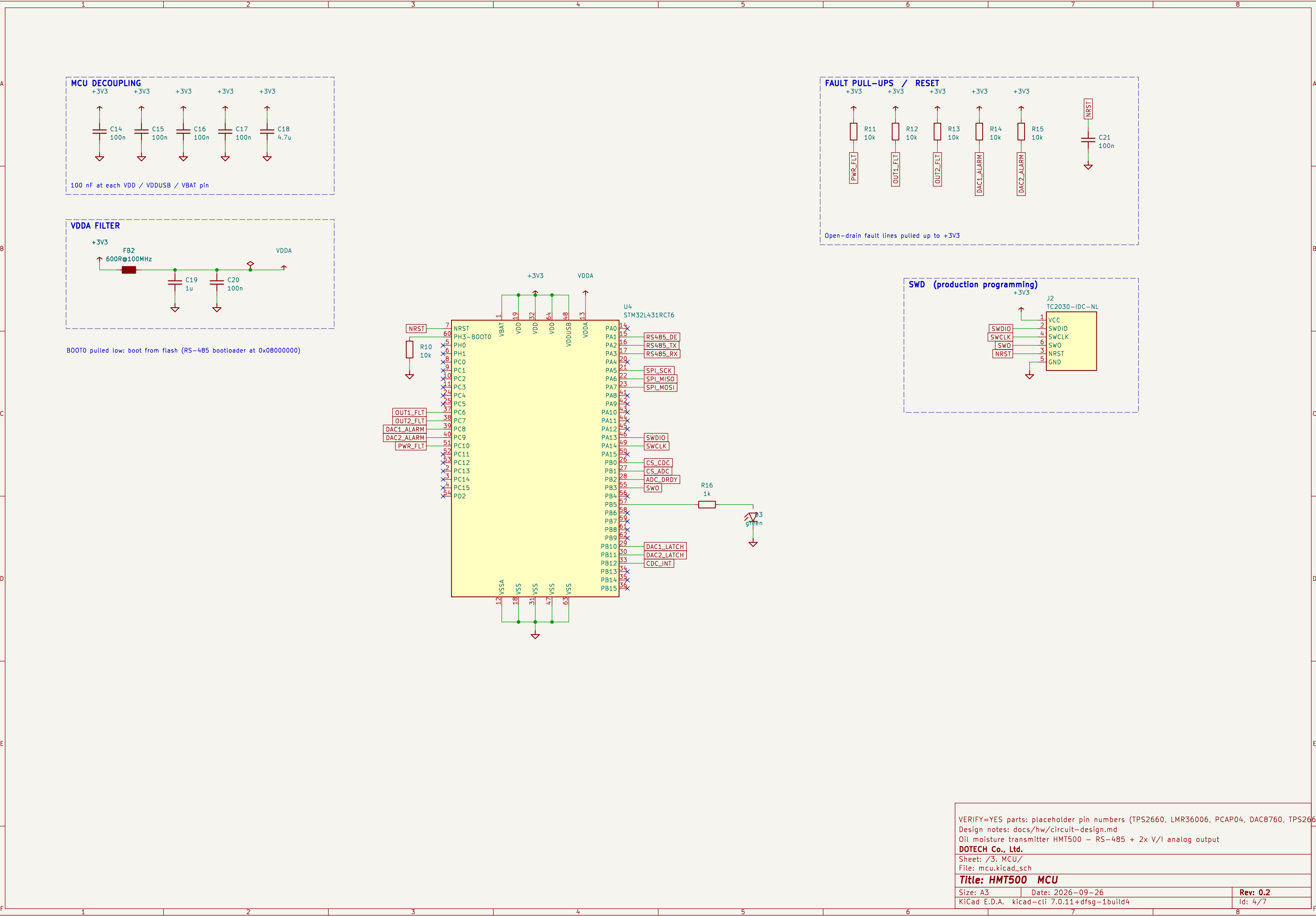
BUCK 5 V LMR36006 (4.2-60 V in, 0.6 A)

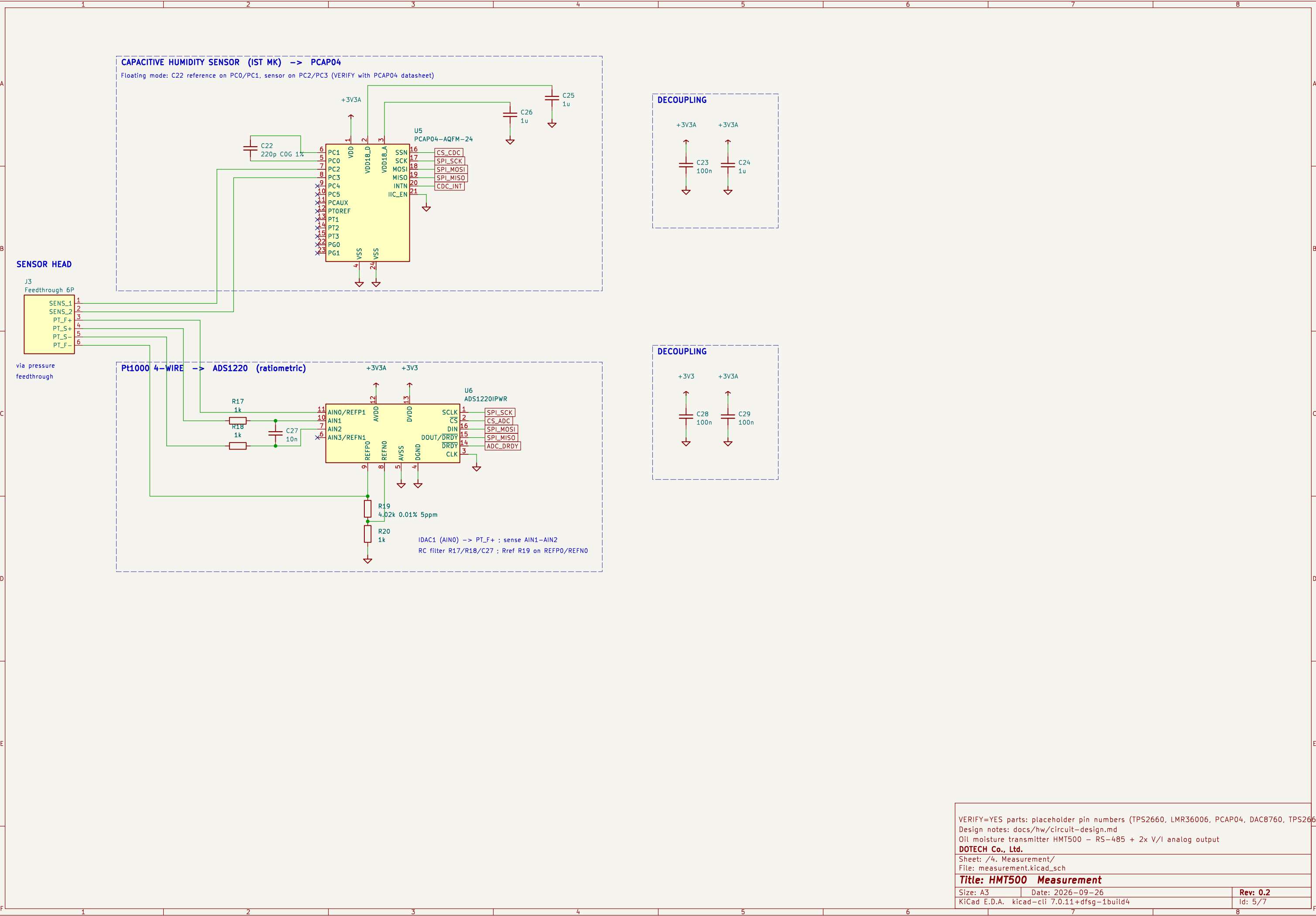


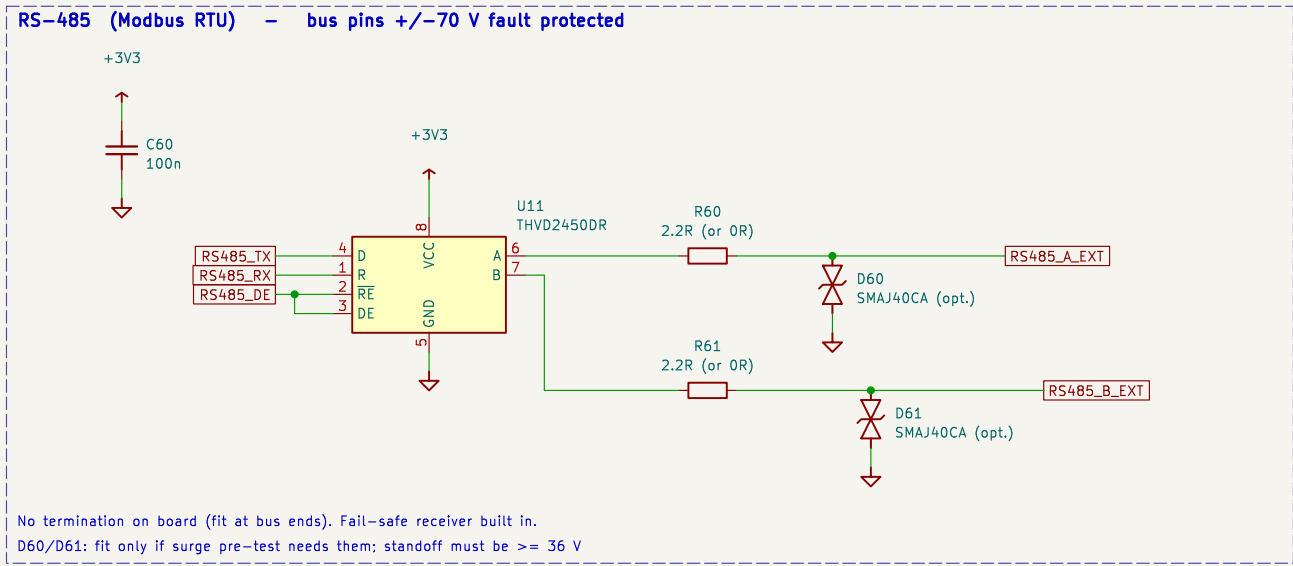
$V_{out} = 1.0\text{ V} \times (1 + R8/R9) = 5.0\text{ V}$ (Vref 1.0 V: VERIFY)

LDO 3.3 V TPS7A2033 + analog rail +3V3A









VERIFY=YES parts: placeholder pin numbers (TPS2660, LMR36006, PCAP04, DAC8760, TPS26611)

Design notes: docs/hw/circuit-design.md

Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

Sheet: /6. RS-485/

File: rs485.kicad_sch

Title: HMT500 RS-485

Size: A4

Date: 2026-09-26

Rev: 0.2

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Id: 7/7